ACE Cyber Project Charter: Fall 2025

1. Student(s) & Team Information

Principal Investigator (PI) Name(s):
University:

_____Denzel Jefferson_____
_____Charleston Southern University_____

2. Faculty / DoD Mentor Information

Faculty Advisor:
DoD/ACE Cyber Mentor (if applicable):

_____Professor Julie Henderson_____
_____Professor William Littleton_____

3. Project Title

Provide a clear, concise title for your project

Contextualized Threat Intelligence Prioritization for CTI Analysts

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4. Problem Statement

Which research project did you select? Why is this important?

The research project I selected is “Contextualized Threat Intelligence Prioritization for CTI Analysts.” This project is important because CTI Analysts are overwhelmed by the large amounts of raw, unfiltered threat intelligence. This excess, unfiltered threat intelligence data limits efficiency, causes significant alert fatigue, and increases the time required to identify the most critical threats. This project aims to reduce alert fatigue and shorten the time required to identify critical threats by using contextualized operational and threat intelligence data._

5. Research Question

Define the primary question your project seeks to answer

How can operational environment data gathered from Elasticsearch be used to automate and improve threat intelligence prioritization within CTI workflows?

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6. Hypothesis (if applicable)

What do you expect to discover or prove?

If operational environment data from Elasticsearch is incorporated into an automated MISP-based (simulated) threat-prioritization system, then the time required for CTI analysts to identify high-priority threats will significantly decrease while maintaining or improving overall prioritization accuracy compared to a manual review process.

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7. Project Objectives

List 3–5 clear objectives for your project — measurable and achievable in ~3 months

Build a working Proof-of-Concept (PoC) demonstrating a basic ranking system for a simulated intelligence prioritization system using Elasticsearch environment data.

Develop a prioritization system to identify threats as High, Medium, or Low Importance.

Provide and present a final documentation report explaining in detail the results, design, and outcome of the project.

Analyze the efficiency of using an automated prioritization system vs manual review._____
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8. Research Method Selection

Circle or highlight one and explain why you chose it:
- Applied Experiment
- Applied Observational Study

I chose an applied experiment because it best fits what I am conducting in this project. I aim to compare how using an automated prioritization system will identify threats with a High, Medium, and Low importance compared to a manual review._____
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9. Scope & Deliverables

What will you produce? What's in/out of scope?
Deliverables:

- A Proof-of-Concept (PoC) with integration of Elasticsearch data.

- Final documentation explaining the final design and results.
- A tested evaluation comparing automated prioritization to manual review.
- An automated prioritization system that will rank security threats with High, Medium, and Low importance

Out-of-Scope Items:

- Real-time deployment within DoD systems.
- Replacing human CTI analysts with fully AI counterparts.
- Large-scale, company production-ready execution.
- Integration within SOC or SIEM systems.

10. Resources & Tools Needed

List technologies, datasets, labs, or access required

Elasticsearch for collecting and analyzing operational environmental data.

Contextual scoring model implemented within an Elasticsearch environment

Access to the Cyber Innovation Range (CIR) environment.

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11. Timeline (3-Month Scope)

Month 1:

- Specify the ranking logic between High / Medium / Low.
- Finish project design and success strategy.
- Begin setting up a test environment.

Month 2:

- Develop the PoC ranking system.
- Start testing prioritization accuracy.
- Compare automated test results to manual test results.

Month 3:

- Prepare documentation and final presentation.
- Deliver the final report and demonstration.
- Finalize PoC and confirm testing results.

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12. Success Criteria & Metrics

How will success be measured? What are the KPIs or evaluation criteria?

My working PoC will accurately align High, Medium, and Low priority threats with simulated threat intelligence prioritization outputs.

The automated ranking system will significantly reduce the time spent to identify high-priority threats compared to manual review.

The prioritization system will show a noticeable difference in alert clutter, which will help CTI analysts reduce alert fatigue.

A working PoC and final report/documentation are completed and presented before the deadline._____
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13. Risks & Mitigations

Identify key risks — technical, data, schedule — and how you plan to address them

Risk: Time constraints within a 4-month window may pose issues for the project.

Mitigation: Prioritize a working PoC first, along with ranking, and then develop anything extra as optional enhancements.

Risk: The automated ranking system may not align with projected human analyst expectations.

Mitigation: Compare automated results with manual results and tune the ranking logic to align them.

Risk: Potentially unreliable CTI data may compromise the project's accuracy.

Mitigation: Use multiple datasets and test among samples to ensure project accuracy.

14. Expected Impact

How does this contribute to cybersecurity research for DoD?

This research supports the DoD's push for automated data-driven cybersecurity by improving how threat intelligence is prioritized. It will help cyber analysts focus on the most relevant threats rather than sifting through large volumes of potential threats, thereby strengthening overall cyber defense effectiveness and network protection._____

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